

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0165

Henry County 230 kV
December 31, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0165. The Interconnection Customer proposes a 333.6 MW solar photovoltaic facility interconnecting at the Henry County 230 kV substation in the Indiana area. The request was received on October 11, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 125 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$20,827,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of E Methuen to Henry County 500.0 kV, the Murdock to Henry County 230.0 kV L loads to 132.2% of its 558 MVA rating. Project DFAX: 59.8%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Murdock to Henry County 230.0 kV L	Rebuild line with higher-capacity conductor	\$22,889,000	91% (\$20,827,000)
	Total allocated to project		\$20,827,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.